

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Operations Research	Course Code:	MT 4031
	Degree Program:	BSCS	Semester:	Fall 2025
	Exam Duration:		Total Marks:	50
	Submission Date:	30-10-2025	Weight	
	Section:	E and F	Page(s):	
	Exam Type:	Assignment-2		

Reference book: Hamdy A. Taha, Operations Research, An Introduction (10th Edition)

Instruction:

- Clearly write your name, roll no, section, Course title and assignment title on the first page.
- Write question no. as given in Assignment.
- Late submission will have no credit.
- The question that you need to be solved without GEOGEBRA, Excel and/or Python, should be solved by hand.
- Python code must be commented and executable.
- Excel Solver screenshots must include constraints, objective, and results.
- Submit your work as a single PDF file.

Questions 1: [10 marks]

For the following linear model, construct dual, solve dual and find optimal primal solution using two different methods.

$$\begin{aligned} \min \quad & z = 6x_1 + 3x_2 \\ \text{subject to} \quad & 6x_1 - 3x_2 + x_3 \geq 25 \\ & 3x_1 + 4x_2 + x_3 \geq 55 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Question 2: [10 marks]

Solve the following linear model using generalized simplex method.

$$\begin{aligned} \text{Maximize } z = & x_1 - 3x_2 \\ & x_1 - x_2 \leq 20 \\ & x_1 + x_2 \geq 40 \\ & 2x_1 - 2x_2 \geq 30 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Question 3: [10]

Consider the following model:

$$\begin{aligned} \text{Maximize } z &= 5x_1 + 12x_2 + 4x_3 \\ x_1 + 2x_2 + x_3 + x_4 &\leq 5 \\ 2x_1 - x_2 + 3x_3 &= 1 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

Identify the best solution from among the following basic feasible solutions. Generate entire simplex tableau in case of.

$$\text{Basic Variable} = (x_2, x_1), \text{Inverse} = \begin{pmatrix} \frac{2}{5} & -\frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{pmatrix}$$

Question 4: [10]

- a. In the unbalanced transportation problem in Table 5.36, if a unit from a source is not shipped out (to any of the destinations), a storage cost is incurred at the rate of \$5, \$4, and \$3 per unit for sources 1, 2, and 3, respectively. Additionally, all the supply at source 2 must be shipped out completely to make room for a new product. Apply **Vogel's approximation** method and find the **optimal solution**.

1\$	2\$	1\$	20
3\$	4\$	R\$	40
2\$	3\$	3\$	30
30	20	20	

Description about **R**: **R is the sum of last four numeric digits of your roll number. For example, if your roll number is 21L-0930 then R= 0 + 9 + 3 + 0 = 12.**

- b. Solve the above problem using **the Least cost method** and find the initial basic feasible solution. Compare the solution, which method provides better initial feasible solution.

Question 5: [10]

The coach of an age group swim team needs to assign swimmers to a 200-yard medley relay team to send to the Junior Olympics. Since most of his best swimmers are very fast in more than one stroke, it is not clear which swimmer should be assigned to each of the four strokes. The five fastest swimmers and the best times (in seconds) they have achieved in each of the strokes (for 50 yards) are

Stroke	Carl	Chris	David	Tony	Ken
Backstroke	37.7	32.9	33.8	37.0	35.4
Breaststroke	43.4	33.1	42.2	34.7	41.8
Butterfly	33.3	28.5	38.9	30.4	33.6
Freestyle	29.2	26.4	29.6	28.5	31.1

The coach wishes to determine how to assign four swimmers to the four different strokes to minimize the sum of the corresponding best times.

- (a) Formulate this problem as an assignment problem.
- (b) Obtain an optimal solution using Hungarian method.
- (c) Solve this problem with **python** to verify the results found in part (b).